

AI-Based Tree Monitoring System

Project Requirements & Technical Design Document

1. Introduction

This document outlines the requirements and proposed technical architecture for an AI-based ecological monitoring system designed to analyze satellite imagery and detect individual trees.

The project utilizes a dataset of approximately 3,200 satellite images. These images are of sufficient resolution to identify individual trees, allowing the system to detect trees and monitor their survival rate and vegetation growth trends over time.

The primary objective is to develop an automated monitoring system that extracts ecological indicators from satellite imagery, helping organizations track reforestation and vegetation development more efficiently.

2. Project Objectives

The main objectives of the system include:

- Detect individual trees from satellite imagery
- Monitor tree survival rates across different time periods
- Estimate vegetation growth trends

- Provide measurable ecological indicators for monitoring projects
- Reduce reliance on manual field monitoring
- Provide scalable analysis across large geographic areas

3. Dataset Overview

The current dataset consists of approximately 3,200 satellite images with sufficient resolution to identify individual trees. These images will be used to train and fine-tune computer vision models capable of detecting and segmenting trees within the imagery. [Dataset source](#).

4. Project Constraints

The capabilities of the system depend heavily on the quality and availability of training data. The current dataset introduces several constraints that influence the design of the system.

- The dataset does not categorize trees by species.
- The system cannot determine the exact biological health of each tree.
- The system cannot determine the specific cause of tree death.
- Satellite imagery may contain noise or environmental interference.

5. Data Sources

The system will rely on the following data sources:

- Satellite imagery (e.g., Sentinel or similar providers)
- Existing monitoring datasets

- Citizen monitoring records and reports

6. System Overview

The proposed solution introduces an AI analysis layer that processes satellite imagery to extract ecological indicators.

System Architecture Components

Component	Function
Data Source Layer	Stores satellite imagery and monitoring datasets
AI Processing Layer	Analyzes imagery and extracts indicators
Application Interface Layer	Provides API services for exposing results

7. System Features

The system is designed with a comprehensive set of features to meet the objectives of automated ecological monitoring, data analysis, and reporting.

Feature Area	Description	Key Capabilities
User & Access Management	Securely manage user roles and access permissions.	Authentication, Role-based user management, Project-specific access control.
Plantation Site Management	Tools for defining, organizing, and tracking plantation projects.	Listing and filtering of plantation sites, Site boundary definition (GIS integration), Historical monitoring data retrieval.
AI-Powered Analysis	Core functionality for processing satellite data and extracting ecological indicators.	Tree detection and counting, Vegetation survival rate calculation, Growth trend estimation, Plantation analysis (density, coverage).
Reporting & Transparency	Generate transparent, auditable reports on ecological impact.	Project monitoring reports, Carbon neutralization calculation for all projects, Transparency check via blockchain audit trail, Data immutability verification.
Data Visualization	Intuitive interface for exploring monitoring data and results.	Interactive maps with tree detection overlays, Vegetation growth dashboards, Time-series analysis charts, Exportable data and reports.

Feature Area	Description	Key Capabilities
System Administration	Management of system health, data sources, and configuration.	Data pipeline monitoring, Model version control, Satellite imagery source integration management.

8. Core Technologies

The system will use stable and widely adopted technologies suitable for both research and production environments.

- **Python:** For backend development and machine learning workflows
- **FastAPI:** For building high-performance APIs
- **PyTorch:** For deep learning and computer vision models

9. Image Analysis Approach

The system will fine-tune the YOLOv26 model, which already provides strong capabilities in object detection and image segmentation. Fine-tuning this model will enable accurate detection of individual trees in satellite imagery.

Key Analysis Functions:

- Detect individual trees

- Identify planting areas
- Measure vegetation coverage
- Track visible vegetation changes over time

10. Data Pipeline

The system will follow a structured data processing workflow:

1. Collect satellite imagery
2. Preprocess images (cropping, normalization, filtering unusable images)
3. Process images using AI models
4. Generate ecological indicators
5. Store results and expose them via an API

11. Frontend System

A web interface will be developed using React. The frontend will provide visualization and monitoring capabilities.

- Interactive maps displaying satellite imagery
- Tree detection overlays
- Vegetation growth dashboards
- Project monitoring reports

12. Blockchain Integration

The system will integrate blockchain technology to ensure transparency, traceability, and integrity of environmental monitoring data.

- Immutable storage of monitoring records
- Transparent verification of tree planting data
- Secure audit trails for environmental reporting
- Proof-of-impact for ecological projects

13. Integration Strategy

The project deployment will follow a phased approach:

Phase	Objective	Key Deliverables	Timeline
Phase 1 – Prototype	Develop AI models and validate indicators.	Trained model, validated ecological indicators.	1 month
Phase 2 – Pilot	Deploy API service and integrate with selected projects.	Functional API, initial project integration.	1 month
Phase 3 – Expansion	Scale to additional regions and improve model accuracy.	Scalable system, improved model	1 month

Phase	Objective	Key Deliverables	Timeline
		performance, wider regional coverage.	

14. Expected Outcomes

- Automated large-scale ecological monitoring
- Consistent analysis across projects
- Reduced manual monitoring workload
- Improved environmental reporting transparency

15. Conclusion

This system provides a scalable and practical approach for monitoring vegetation and tree growth using satellite imagery and artificial intelligence. By combining machine learning, satellite data, blockchain transparency, and modern web technologies, the platform enables reliable environmental monitoring and reporting.

16. Roles and Responsibilities

The successful execution of this project relies on a clear definition of roles and responsibilities within the development team.

Role	Responsibility	Team Member
Project Owner	Overall project direction, scope, and strategic decision-making.	Asawir Khokhar
Product Owner/Lead	Overall product strategy, roadmap, feature prioritization, stakeholder communication, and task assignments	Asawir Khokhar
AI Engineer	Development, training, and fine-tuning of the computer vision models (e.g., YOLOv26) for tree detection and segmentation.	Asawir Khokhar
Frontend Developer	Development of the web interface (React), including interactive maps, dashboards, and reporting views.	Muhammad Usama Iqbal
Backend Developer	Development of the API services (FastAPI), data pipeline logic, and integration with the AI Processing Layer.	Huzaifa Rathore
Blockchain Developer	Implementation and maintenance of the blockchain integration for	Muhammad Usama Iqbal, Huzaifa Rathore

Role	Responsibility	Team Member
	data immutability, transparency, and audit trails.	